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RESEARCH ARTICLE

Generating Powerful Encryption Keys for Image Cryptography With Chaotic Maps by Incorporating Collatz Conjecture

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ABSTRACT In recent years, the chaotic logistic map key (CLMK) has been widely used for image encryption due to its ability to generate random and unpredictable sequences of numbers. The CLMK algorithm uses the chaotic logistic map to generate a key stream, which is then used to encrypt the plaintext image. This study proposes an image encryption scheme based on CLMK, Collatz Conjecture, and indexing techniques; it uses a private key of a 15-digit size that is parsed in a novel way to generate a considerable keyspace reaches up to 2^{97}. The Collatz Conjecture generator is used to create variable length sequences that are used to generate the Chaotic parameters r_1, r_2, x_1, and x_2; these parameters are used to create a 2D CLMK for encrypting each color channel and a 1D vector that is used for permeation, both confusion and diffusion are performed in four consecutive rounds to ensure the robustness of the proposed approach. The random sequences generated by the Collatz Conjecture were tested for randomness using the NIST benchmark randomness test suit, and they passed most tests. The proposed method is robust against all the well-known attacks, including differential attacks, white Gaussian noise attacks, Salt-and-pepper noise attacks, clipping attacks, occlusion attacks, and known plaintext attacks. Various experiments were conducted to evaluate the performance of the proposed method in terms of encryption time, throughput, and several image quality metrics. The results indicate that the proposed method is efficient regarding encryption time, and throughput enhancement can reach up to 3.34 times the state-of-the-art chaotic encryption methods.

INDEX TERMS Chaotic logistic map, chaotic logistic map key, chaotic parameter, Collatz conjecture, cryptography.

I. INTRODUCTION

Digital images, from personal photos to medical imaging and scientific research, have become essential to our daily lives. However, with the widespread use of digital images comes the need for robust security measures to protect sensitive information and prevent unauthorized access and tampering [1], [2], [3]. Figure 1 shows an example of colored image cryptography.

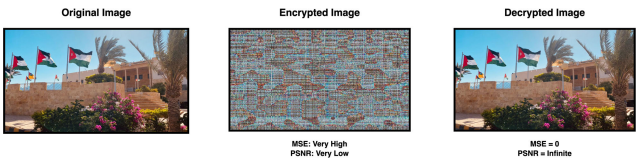


FIGURE 1. Color image cryptography.

Image encryption is a crucial area of research that focuses on securing digital images using cryptographic techniques. Private key cryptography is one of the primary methods used in image encryption, where a single key is used to encrypt

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and decrypt the data, as shown in Figure 2. However, with the increasing complexity of encryption algorithms and the emergence of new threats, it has become necessary to develop more advanced encryption techniques [4], [5].

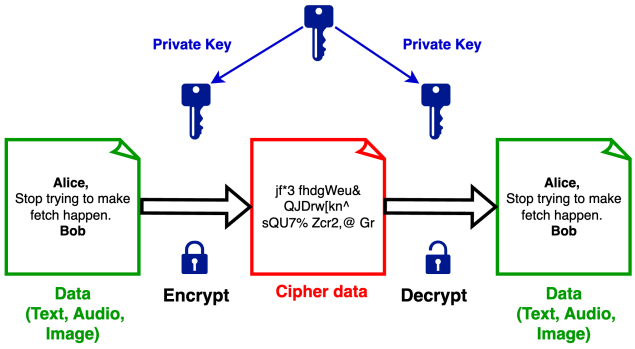


FIGURE 2. Private key cryptography.

One promising area of research in image encryption is the application of chaos theory. Chaos theory is a branch of mathematics that studies the behavior of nonlinear systems susceptible to initial conditions. Using chaos theory in image encryption can enhance the security of private key cryptography algorithms by introducing a new layer of randomness and unpredictability to the encryption process [6], [7], [8].

Integrating chaos theory into image encryption and private key cryptography can significantly enhance digital image security and protect sensitive information from unauthorized access and tampering. Chaos-based image cryptosystems are encryption techniques that use chaotic dynamics to secure digital images. These systems work by transforming the image into an unrecognizable form through the use of two phases: confusion and diffusion. In the confusion phase, pixel positions are rearranged throughout the image while keeping their values constant. This process renders the image unintelligible and serves as the first line of defense against unauthorized access. However, as this phase alone is not secure enough, the diffusion phase is added to change the pixel values in sequence with the help of a chaotic map. This process increases the level of security of the encrypted image. Chaos-based image cryptosystems have been used in various fields, such as military and medical imaging, to protect sensitive information from unauthorized access [7], [8], [9], [10], [11].

This research paper aims at reducing the private key size while still achieving the same strength of encryption on images as other state-of-the-art encryption algorithms, incorporating Collatz Conjecture mathematics and indexing techniques to parse the small private key in a novel way that achieves a huge keyspace-moreover, proofing the randomness of the generated sequences using the NIST randomness benchmark test suit. The main contribution of this study is the proposal of a small private key and yet achieving huge key space through the incorporation of the Collatz Conjecture to parse and generate random sequences. Moreover, intensive experiments are conducted to prove the robustness of the proposed method against the most famous attacks.

The organization of this paper unfolds in the following manner: Section II delves into the literature associated with the suggested approach, while a comprehensive examination of the proposed framework is presented in Section III. Section IV reveals the encryption procedure's implementation specifics. The effectiveness of the proposed encryption method is demonstrated in V, and finally, Section VII outlines the conclusions reached and suggests some future directions to extend the suggested approach further.

II. RELATED WORK

An encryption algorithm that is considered best takes less computational time without compromising security.

The one-time pad (OTP), introduced by Frank Miller, is a cryptographic technique that provides perfect secrecy for encrypting messages [12], [13]. It is an encryption method that uses a random key, known as the one-time pad **, which is as long as the message is encrypted**. The key is used only once and never reused for any other message. The security of the one-time pad lies in the randomness and secrecy of the key [14]. However, the practical implementation of the one-time pad can be challenging [15]. The key must be securely generated, distributed, and stored, and some research has been made to address this [16], [17]. It **must be at least as long as the plaintext**, and the key must remain secret and never be reused. If any of these requirements are compromised, the encryption's security can also be compromised.

The length of the key is problematic, especially when the key itself has to be transmitted over an insecure channel, as any secure method of transmitting it would be equivocally a method that could be used to send the message itself.

The other thing is that we rely on the generation of random numbers. A lot of work in this area has been published by various scientists, including Sadeghiyan and Pieprzyk [18], [19] and Uschner et al. [20].

The essence of sending a much ''shorter key'' that is secure is paramount. This is especially true when one is dealing with images and videos. For example, many videos are currently being taken that may require encryption, such as security videos in schools and even private homes. The surveillance webcams, for instance, at a Cape Breton school in Nova Scotia, Canada, hacked into and publicly displayed [21]. Thus, fast encryption may be needed to safeguard against this. A one-hour HD video, for example, requires 500 GB of storage [22]. Having a one-time pad of 500 GB to store and transmit would not only double the size of the data being stored but would be impractical to transmit.

When dealing with chaotic maps, a well-established random number generator must generate the **seed**. This is then used to produce both r and x. Thus, the concept of the chaotic map need not enjoy the property of indistinguishability from a truly random generator [23]. Still, as accepted in

the cryptographic community, when using chaotic maps with images, they satisfiably pass the NIST random test-suite [24], [25].

Chaos-based cryptographic methods provide several advantages, such as increased flexibility, high security, less computational overheads, less computing power, and ease of implementation [8], [11]. Also, in most chaos-based encryption schemes, confusion and diffusion of the encryption key are performed at the pixel level. These characteristics make them a promising alternative to conventional cryptographic algorithms for encrypting and decrypting a wide range of digital data. Additionally, various chaotic maps can be incorporated into chaos-based cryptosystems, increasing the options. Thus, chaotic cryptography has gained more attention than conventional methods because of its lower mathematical complexity and better security [26], [27], [28].

The architecture of a chaos-based image cryptosystem can be divided into two distinct phases: confusion and diffusion. Figure 3 depicts a block diagram of this architecture. The confusion phase, also called pixel permutation, rearranges the positions of pixels throughout the entire image without changing their values, rendering the image unrecognizable. However, this phase alone is not secure enough and can be easily hacked by attackers. Therefore, the diffusion phase is necessary, which modifies the pixel values of the entire image in sequence with the help of a chaotic map. The confusion-diffusion process is repeated several times until an acceptable level of security is achieved [8], [28]. In the proposed method, the confusion-diffusion process is applied in four consecutive rounds to ensure full randomness so that any spatial correlation among neighbor pixels vanishes. To further ensure the randomness of the confusion-diffusion process, the keys are random and generated using the Collatz Conjecture; further details are provided in Section III.

To enhance the security of the encrypted image, both the confusion and diffusion processes should be iterated more than once. This approach is supported by research published in [29], where the authors found that a one-round Substitution Permutation Network-based cryptosystem was not secure enough to avoid security problems, even when a chaotic system was used as the core of the encryption system. In [28], the authors used four rounds of confusion and diffusion processes to enhance the security of their encryption system. Overall, the use of multiple rounds of these processes can significantly improve the security of encrypted data.

There are several chaotic maps used in image encryption. No single ''best'' chaotic map is used in image encryption, as the choice depends on the specific requirements and constraints of the application in question. A comparison of some popular chaotic maps used for image encryption is shown in Table 1:

In [31], the authors compared the performance of various encryption algorithms for digital images, including the chaotic map-based algorithm. Their study found that the chaotic map-based algorithm provided a higher level of security than other algorithms, such as AES and RSA. However, the throughput analysis of the algorithm was not reported; a summarization of their findings is shown in Table 2.

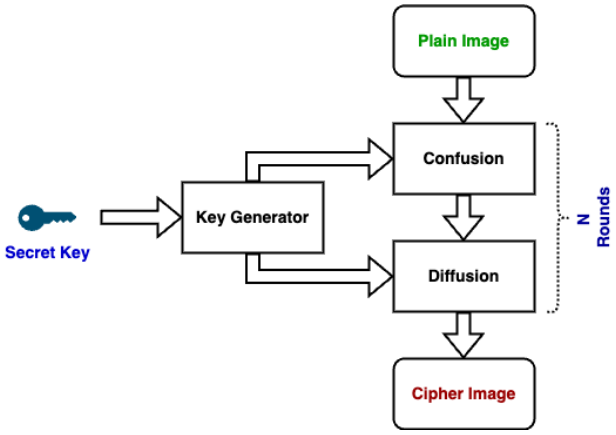


FIGURE 3. Chaos-based image cryptosystem architecture.

Hua et al. in [32], proposed an image encryption algorithm based on a cosine-transform-based chaotic system. The algorithm utilized a three-dimensional chaotic map to produce a chaotic image-encryption sequence. Their experimental results showed that the proposed algorithm performed better than existing algorithms. The throughput analysis was not reported in the paper.

The authors in [33] presented a novel image encryption algorithm based on a polynomial combination of chaotic maps and dynamic function generation. The algorithm combined chaotic maps and dynamic function generation to encrypt images. Their experimental results showed that the proposed algorithm performed better than existing algorithms. The throughput analysis was not reported in the paper. In [34], the authors proposed a multiple-image encryption algorithm based on DNA encoding and a chaotic system. The algorithm utilized chaotic maps and DNA encoding to encrypt multiple images.

The authors in [35] proposed a multiple-image encryption scheme with bit-plane decomposition and chaotic maps. The algorithm utilized bit-plane decomposition and chaotic maps to encrypt multiple images.

Heucheun Yepdia et al. in [36], presented a robust and fast image encryption scheme based on a mixing technique. The algorithm utilized a chaotic map and a mixing technique to encrypt images.

In [37], Zhang et al. proposed a novel image encryption algorithm that utilized the chaotic logistic map to generate random numbers for encryption.

Another study by Liu et al. in [38] proposed an image encryption algorithm based on a chaotic logistic map and a hyperchaotic Lorenz system. The proposed method used a combination of these chaotic systems to generate the key stream for encryption, improving the encrypted image's

TABLE 1. Comparisons of some popular chaotic maps used for image encryption [6], [11], [30].

Chaotic Map	Dimensions	Sensitivity to Initial Conditions and Parameters	Nonlinearity and Complexity	Computational Efficiency	Robustness against Attacks
Logistic map	1D	High	Moderate	High	Moderate
Arnold's cat map	2D	High	Moderate	High	Moderate
Hénon map	2D	High	High	Moderate	Moderate
Tent map	1D	High	Moderate	High	Moderate
Ikeda map	2D	High	High	Moderate	Moderate
Julia sets	2D	High	High	Low	Moderate to Low
Mandelbrot sets	2D	High	High	Low	Moderate to Low
Lorenz system	3D	High	High	Moderate to Low	Moderate

TABLE 2. Performance comparisons of different cryptography methods [31].

Method	Throughput (Kbytes per second)
Non-chaotic approach	170.3906
Chaotic approach	141.2305
Hyper Chaotic approach	636.3379

TABLE 3. Performance comparisons of methods mentioned in [36] and [28].

Method	Throughput (Kbps)
Yepioda et al. [36]	888.8867
Hua et al. [32]	638.4082
Asgari-chenaphlu [33]	911.0352
Zhang and Wang [34]	360.4102
Zhenjun and Sun [35]	384.9609
Abu-Faraj et al. [28]	302.7968

security. A study by Lu et al. in [39] proposed a hybrid image encryption algorithm that combined the chaotic logistic map and the Arnold transform.

In [40], Hu et al. proposed an image encryption algorithm that utilized the chaotic logistic map and the DNA encoding algorithm. Finally, a study by Li et al. in [41] proposed an image encryption algorithm that utilized a 3D chaotic map and the logistic map to generate the key stream for encryption.

In [28], Abu-Faraj et al. proposed a color image cryptography technique using a chaotic logistic map model to generate two complex secret keys for encryption and decryption. The model generates a 2D key by incorporating matrix dimensions R and C. The generated pseudo-random numbers were acceptable and comparable to existing literature. The method offers high security due to its sensitivity to minor key changes and a large key space, reducing the likelihood of hacking. Performance analysis revealed that the proposed technique improves image cryptography by reducing encryption-decryption time and increasing throughput, outperforming existing methods.

In conclusion, the use of chaotic maps in image encryption has been extensively studied, and several algorithms have been proposed to improve the security and throughput of the encryption process. The throughput analysis of some of these algorithms shows that they perform better than existing algorithms. Table 3 shows the performance comparisons of the different methods studied in [32], [33], and [34].

The Collatz Conjecture, or the $3n + 1$ conjecture, is a famous unsolved mathematics problem comprising a simple iterative algorithm applied to positive integers. The conjecture posits that for any positive integer, if it is even, divide it by 2, and if it is odd, multiply it by 3, add 1, and continue the process with the resulting number. The conjecture states that no matter what positive integer you start with, this process will eventually reach number 1. The Collatz conjecture can be expressed according to the Equation (1) [42]:

$$f(n) = \begin{cases} \dfrac{n}{2}, & \text{if } (n \bmod 2 \equiv 0) \\ 3n + 1, & \text{if } (n \bmod 2 \equiv 1\ ), \end{cases} \tag{1}$$

where n is a positive integer, and $f(n)$ is the function that generates the next term in the sequence. The Collatz Conjecture states that no matter what positive integer n you start with if you repeatedly apply the function $f(n)$, the sequence will eventually reach the value 1.

In [43], the authors proposed a novel image encryption scheme based on the Collatz conjecture. In this scheme, the plaintext image is converted into a binary sequence and processed using the Collatz conjecture to generate a pseudo-random sequence. This sequence is then used to encrypt the image using an XOR operation. The authors claim that this scheme provides good security and resistance to attacks.

The authors in [44] proposed using visual cryptography in software watermarking. Visual cryptography is a technique that allows the encryption of an image into several shares, which are then distributed to different parties. This allows the image to be reconstructed only when a sufficient number of shares are combined. The authors suggest using this technique to embed a watermark into software code. In [45], the authors proposed an LSB-based random embedding image steganography technique using modified Collatz conjecture. In this technique, the plaintext image is converted into a binary sequence, and then the LSBs of the pixels are changed using the Collatz sequence. The authors claim that this technique provides good security and imperceptibility.

In summary, the Collatz conjecture has been used in various ways in image encryption and steganography techniques, including generating pseudo-random sequences, modifying LSBs, and creating shares for visual cryptography. These techniques aim to provide good security and resistance to attacks while preserving the quality and integrity of the original image.

The other thing is that we rely on the generation of random numbers. A lot of work in this area has been published by various scientists, including [46]

The essence of sending a much ''shorter key'' that is secure is paramount. This is especially true when one is dealing with images and videos.

When dealing with chaotic maps, a well-established random number generator must generate the **seed**. This is then used to produce both r and x. Thus, the concept of the chaotic map need not enjoy the property of indistinguishability from a truly random generator. Still, as accepted in the cryptographic community, it suffices that they satisfactorily pass the IEEE random test-suite [47].

A highlight of logistic maps' limitations in terms of complexity and throughput performance is presented in this section, so our presented encryption scheme is trying to mitigate these limitations by employing a novel way of parsing private key to generate random sequences through incorporating Collatz Conjecture so that it enhances the nonlinearity and complexity of the proposed method and improves its throughput.

III. THE PROPOSED SCHEME

In this research paper, we will introduce a method of image cryptography; this method will meet all the requirements of data cryptography, which must satisfy the following requirements, especially [48], [49], [50], [51]:

- Strong encryption key: The encryption method must have a strong and complicated key that unauthorized users cannot easily guess or crack.
- Sensitivity to slight changes: The encryption method should be highly sensitive to the slightest changes in the input image or encryption key. This ensures that any attempt to modify the image or the key results in a completely different encrypted image.
- Good quality parameters: The encryption method should provide good-quality parameter values during the encryption and decryption phase. This ensures that the decrypted image is high quality and identical to the original image.
- High MSE during encryption: The Mean Squared Error (MSE) should be high, meaning the encrypted image appears distorted and difficult to understand.
- Zero MSE during decryption: The MSE should be zero, meaning the decrypted image should be identical to the original image.
- Fast encryption and decryption: The encryption and decryption times should be minimized, while the encryption and decryption throughput (Bytes encrypted or decrypted per second) should be maximized.
- Compatibility with any image: The encryption method should be easy to use and process any image without causing any change in the methods implemented. This ensures that the method can be used with different types of images without any compatibility issues.

The chaotic logistic map (CLM) is a mathematical model that demonstrates chaotic behavior. It is a discrete-time dynamical system defined by the following recursive Equation (2) [33]:

$$X_{n+1} = rX_n(X_n - 1) \tag{2}$$

where X_n is the system's value at time step n, r is a parameter that determines the system's behavior, and x is the initial value of the system. The value of r determines the behavior of the CLM. The system converges to a single point when r is less than 1. The system exhibits periodic behavior when r is between 1 and 3. When r is between 3 and 3.57, the system exhibits chaotic behavior. When r is greater than 3.57, the system exhibits divergent behavior.

Based on applying the CLM model using the parameters r and x, the R and C parameters are used to set the dimension of the key. Equation (3) describes how the 2D CLM map is generated:

$$CLM(i, j) = X_{n+1} = rX_n(X_n - 1) \tag{3}$$

where CLM is a 2D map of size $R \times C$, $i \leq R$ is the row index, and $j \leq C$ is the column index. Depending on the key size, the 2D chaotic key requires varying times for generation.

To encrypt a digital image using the CLMK, we can use a bitwise XOR operation between the key and the image. The bitwise XOR operation flips the bits in the image for which the corresponding bit is 1 and leaves the bits unchanged for which the corresponding bit in the key is 0. This process is reversible by performing the bitwise XOR operation between the encrypted image and the key.

To decrypt the encrypted image, the recipient needs to have access to the same key. They can perform the same bitwise XOR operation between the encrypted image and the key to recover the original image. It is essential to keep the key secure, as an attacker who knows the key can easily decrypt the encrypted image.

Figure 4 depicts the proposed encryption method, consisting of three main parts: private key parser, confusion, and diffusion. The private key parser parses the 15-digit private key using the Collatz Conjecture to generate the Chaotic parameters. The confusion part is responsible for the permutation of the plain image row-wise and column-wise using a 1D Chaotic vector generated using the private key parser. The Diffusion part performs the encryption by XORing bitwise each permuted color channel with the 2D Chaotic logistic key generated using the private key parser. The operations of confusion and diffusion are performed four times to ensure that the proposed encryption method is robust against known-plaintext attacks. More details about the key

parser, the encryption, and the decryption phases are provided in the following subsections.

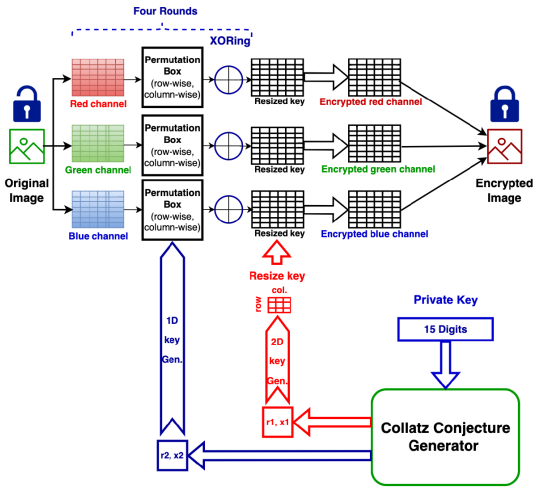


FIGURE 4. The proposed method.

IV. CHAOTIC PARAMETERS GENERATION AND ENCRYPTION/DECRYPTION PHASES

A. KEY GENERATION AND COMPUTATION OF r AND x

The essence of this paper is to use a small key that can produce chaotic maps of similar strengths to state-of-the-art algorithms. A 15-digit random number is generated and partitioned as seen in Table 4 below, such that t is 7 digits long, p and q are 3 digits long, and s is 2 digits long.

TABLE 4. Private key structure.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
t							p			q			s	

Thus, the number can easily fit into the Mantissa of a "double" data type, as $\lg 10^{15} \approx 50$.

1 Bit Sign	11 bits Exponent	52 bits Mantissa

The parser takes the 15-digit private key as input from Table 4 and generates the chaotic parameters: r_1, r_2, x_1, and x_2, in the following way. The 15 digits easily fit in the Mantissa. Thus, we only need 64 bits.

First, t generates numbers using the Collatz Conjecture (CC) by taking the *least significant decimal* in each iteration and appending it to the already formed number. That is, if the original number is $t = N_0$, and the produced number at each iteration of the CC is N_i, then:

$X_0 = N_0 \pmod{10}$, and
$X_i = 10X_{i-1} + (N_i \pmod{10})$.

Example 1: Let $N_0 = 1234217$

$$
\begin{aligned}
X_3 &= 7263 \ldots \\
N_0 &= 1234217, \\
N_1 &= 3(1234217) + 1 = 3702652, \\
N_2 &= 3702652/2 = 1851326, \\
N_3 &= 1851326/2 = 925663,
\end{aligned}
$$

Thus

To generate a key, r_i and x_i, where $i \in \{1, 2\}$, we use the following parameters:

$r_1 \leftarrow t, p$	$r_2 \leftarrow t, q$
$x_1 \leftarrow t+q, p-s$	$x_2 \leftarrow t+p, p+s$

The first parameter is the CC sequence (X_i) generator, as described above. The second parameter dictates the index of the first digit of the 15-digit sequence being taken to generate the Chaotic map; thereby, the second parameter serves as the page index in the generated random sequence.

Example 2: For example, for simplicity, if we are to consider a 3 sequence number instead of 15, with the following table:

t	p	q	s
217	8	5	3

then, the Collatz Conjecture sequence of 15 would be as follows:

	1	2	3	4	5	6	7	8	9	10	11	
t	217	652	326	163	490	245	736	368	184	92	46	...
$t+q$	222	111	334	167	502	251	754	377	1132	566	283	...
$t+p$	225	676	338	169	508	254	127	382	191	574	287	...

Thus, the values of r_i's and x_i's will be as follows:

	Value
$r_1 \leftarrow t, p$	842
$r_2 \leftarrow t, q$	056
$x_1 \leftarrow t+q, p-s$	214
$x_2 \leftarrow t+p, p+s$	721

Algorithm 1 represents the key parser scheme used to generate the needed chaotic parameters for encryption. The parameters r_1 and x_1 generate the 2D logistic key, as in Equation (3). While the parameters r_2 and x_2 are used to generate a 1D vector for permutation as depicted in Equation (2).

B. ENCRYPTION PHASE

The encryption phase can be performed by applying Algorithm 2. It entails two operations performed in four consecutive rounds: the permutation and the diffusion, as shown in Figure 4. A 1D vector generated using Equation (2) shuffles each color channel's rows and columns in the permutation process. While in the diffusion process, the values of the pixels are updated by bitwise Xoring the original value of the pixel by the corresponding value of the resized 2D chaotic logistic key that is generated using Equation (3).

Algorithm 1 Chaotic Parameters Generation Algorithm

Inputs: *Private key of 15 decimal digits*

Output: *Chaotic Logistic Map Parameters: (r_1, r_2, x_1 and x_2)*

1) *Get the private key to be used for encryption.*
2) *Extract the Collatz Conjecture related parameters: t, p, q, and s.*
3) *Run t through the Collatz Conjecture to generate a random sequence.*
4) *Start at index p in the generated random sequence and take 15 consecutive elements.*
5) *Extract the LSD of each element in the sequence, combine all the digits into one decimal number, and divide it by* 1e15. *This will yield* r_1'.
6) *Run* $t+q$ *through the Collatz Conjecture to generate a random sequence.*
7) *Start at index* $p-s$ *in the generated random sequence and take 15 consecutive elements.*
8) *Extract the LSD of each element in the sequence, combine all the digits into one decimal number, and divide it by* 1e15. *This will yield* x_1.
9) *Run t through the Collatz Conjecture to generate a random sequence.*
10) *Start at index q in the generated random sequence and take 15 consecutive elements.*
11) *Extract the LSD of each element in the sequence, combine all the digits into one decimal number, and divide it by* 1e15. *This will yield* r_2'.
12) *Run* $t+p$ *through the Collatz Conjecture to generate a random sequence.*
13) *Start at index* $p+s$ *in the generated random sequence and take 15 consecutive elements.*
14) *Extract the LSD of each element in the sequence, combine all the digits into one decimal number, and divide it by* 1e15. *This will yield* x_2.
15) *Map the generated* r_1' *to the range 3.57 to 4.00 to ensure the Chaotic behavior, yields* r_1.
16) *Map the generated* r_2' *to the range 3.57 to 4.00 to ensure the Chaotic behavior, yields* r_2.

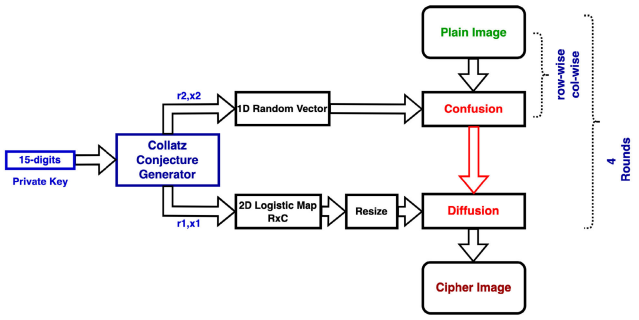


FIGURE 5. Encryption phase.

Figure 5 illustrates the encryption phase of the proposed method, which uses a 15-digit private key in conjunction with the Collatz Conjecture generator, 2D logistic map, and 1D random vector for the confusion and diffusion processes. The process is outlined in the following steps:

- A 15-digit private key is applied to the Collatz Conjecture generator. This generates four variables: r_1, r_2, x_1, and x_2.
- The variables r_1 and x_1 are fed into the 2D logistic map with dimensions R and C. This map is used for the diffusion process, which aims to spread the influence of individual plaintext elements over the entire cipher image.
- In parallel, the variables r_2 and x_2 are fed into the 1D random vector, which is used for the confusion process. This process aims to obscure the relationship between the plaintext and the cipher image, making it difficult to extract information from the cipher without the proper key.
- Both the confusion and diffusion processes are iterated four times, enhancing the security of the encrypted image. The security of an encrypted image can be enhanced by iterating both the confusion and diffusion processes more than once. Several studies in have validated this technique [28], [29]. In [29], the authors analyzed a chaotic cryptosystem that used a one-round Substitution Permutation Network. They showed that even including a chaotic system as the core of the encryption system did not prevent security problems. Therefore, using the confusion and diffusion processes four times can add an extra layer of security to the encryption process. This approach has been

Algorithm 2 Encryption Algorithm

Inputs:

Color image to be encrypted, Chaotic parameters: r_1,r_2, x_1 and x_2, and size of Logistic map ($R \times C$)

Output: *Encrypted image*

1) *Retrieve the image size to be encrypted.*
2) *Extract each color matrix.*
3) for $k = 1$, $k{+}{+}$, while $k <= 4$
   Get a 1D Chaotic vector using r_2 & x_2 input parameters - as random index for row shuffling
   Shuffle the rows
   Get a 1D Chaotic vector using r_2 & x_2 parameters - as random index for column shuffling
   Shuffle the columns
   Use r_1, x_1 and size of Logistic map input parameters to generate a 2D Chaotic Logistic map key, R and C are used to apply two loops to create a 2D matrix. Use r_1, x_1 to calculate each element in the matrix.
   Resize the generated 2D Chaotic Logistic map key to match the size of the color matrix.
   Apply XORing of the resized key with the associated color matrix.
4) *End for*
5) *Combine the encrypted color matrices in one matrix to get the encrypted color image.*

implemented in previous works, such as the one found in [28], where the authors used four rounds of the confusion/diffusion process.
- As shown in Figure 5, the input to the Confusion/Diffusion process is the Plain image, which is the original, unencrypted image. After the encryption process, the output is the cipher image - a securely encrypted version of the original image.

In summary, Figure 5 depicts the proposed encryption phase, which combines the Collatz Conjecture generator, 2D logistic map, and 1D random vector to create a secure cipher image through iterative confusion and diffusion processes.

C. DECRYPTION PHASE

Algorithm 3 explains the steps of the decryption phase. Like the encryption phase, four consecutive rounds of reshuffling and diffusion are performed to restore the original value of pixels in each color channel, as depicted in Figure 4. Any slight change in the chaotic parameters, which are (r_1, r_2, x_1, and x_2), will yield a corrupted decrypted image because the generated keys are sensitive to initial conditions as it will be proved later. Thus, this is considered a hacking attempt.

Algorithm 3 Decryption Algorithm

Inputs:

Encrypted color image, Chaotic parameters: r_1,r_2, x_1 and x_2, and size of Logistic map ($R \times C$).

Output:

Decrypted image

Process

1) *Retrieve the size of the image to be decrypted.*
2) *Extract each color matrix*
3) for $k = 1, k{+}{+}$, while $k <= 4$
   Get a 1D Chaotic vector using r_2 & x_2 input parameters - as random index for row shuffling
   Reshuffle the rows
   Get a 1D Chaotic vector using r_2 & x_2 parameters - as random index for column shuffling
   Reshuffle the columns
   Use r_1, x_1 and size of Logistic map input parameters to generate a 2D Chaotic Logistic map key, R and C are used to apply two loops to create a 2D matrix. Use r_1, x_1 to calculate each element in the matrix.
   Resize the generated 2D Chaotic Logistic map key to match the size of the color matrix.
   Apply XORing of the resized sub key with the associated color matrix.
4) *End for*
5) *Combine the decrypted color matrices in one matrix to get the decrypted color image.*

Figure 6 illustrates the decryption phase of the proposed method, which begins with applying a 15-digit private key to the Collatz Conjecture generator. This generator produces four values: r_1, r_2, x_1, and x_2. These values are then utilized in the subsequent confusion and diffusion processes to transform the cipher image into the plain image.

The r_1 and x_1 values generated by the Collatz Conjecture are fed into a 2D logistic map with dimensions R (rows) and C (columns). This map is used in the diffusion process, which disperses the image data throughout the plain image. On the other hand, the r_2 and x_2 values are input into a 1D random vector, which is utilized in the confusion process. This process obscures the relationship between the plain and cipher images by altering the image data's ordering.

The confusion and diffusion processes are applied iteratively, with four repetitions, to enhance the security of the decryption method. As Figure 6 indicates, the input to these combined processes is the cipher image, the encrypted version of the original plain image. The output of the confusion and diffusion processes is the decrypted plain image, which is the intended final result of the decryption phase.

In summary, Figure 6 demonstrates the decryption phase of the proposed method, which involves the application of a 15-digit private key to the Collatz Conjecture generator and subsequent use of the generated values in the confusion and diffusion processes. These processes are iterated four times to provide a secure decryption method that effectively transforms the cipher image into the original plain image.

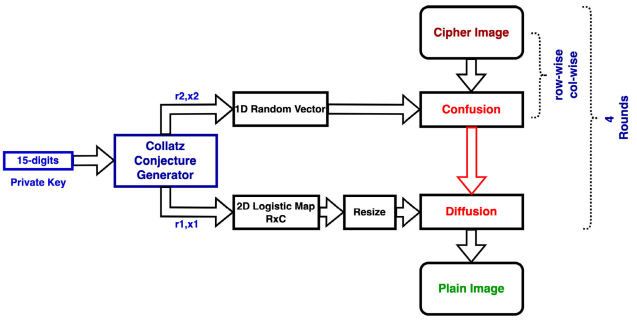


FIGURE 6. Decryption phase.

V. SECURITY ANALYSIS

This article employs a simulation assessment on macOS Ventura 13.3.1 OS, featuring the Apple M1 Pro processor and 16 GB of RAM. The algorithm is developed using MATLAB R2023a software. To test the proposed method, we used a familiar image dataset [52] to obtain random images for the proposed encryption-decryption method. The analysis involved comparing the plots and histograms of the source (original), encrypted, and decrypted images. The chosen photos were encrypted and decrypted using the suggested approach to conduct a performance evaluation and guarantee that other researchers could replicate the outcomes.

The dataset used in this article is Kodak Lossless True Color Image Suite, a comprehensive collection of high-quality, true-color images that serve as a valuable resource for researchers and developers in image processing, compression, and encryption. This image suite, accessible

at [52], provides a consistent benchmark for evaluating and comparing the performance of various algorithms and techniques in image processing.

The imaging suite comprises 24 true color images captured using Kodak's high-quality imaging equipment. These images cover many subjects, including portraits, landscapes, and still-life compositions. The diverse content of the imaging suite allows researchers to test their algorithms and techniques against various image characteristics, such as color distribution, texture, and detail.

A. KEY RANDOMNESS ANALYSIS

Key randomness analysis is a crucial aspect of evaluating the security and efficacy of cryptographic systems. To ensure the robustness of these systems, two predominant methods of randomness testing are employed: the NIST (National Institute of Standards and Technology) randomness test suite and Matlab randomness tests. In the following subsections, we will delve into the results generated by these two approaches. First, the NIST randomness test suite, a comprehensive set of statistical tests designed to assess the quality of random sequences generated by cryptographic algorithms, will be examined [53], [54], [55]. Next, we will explore Matlab randomness tests, which utilize various statistical tools and functions available in the Matlab software environment to scrutinize the randomness of key sequences. By analyzing the outcomes of these two subsections, we can gain valuable insights into the effectiveness and security of the tested cryptographic systems [56], [57], [58].

1) NIST RANDOMNESS TEST RESULTS

The NIST Randomness Test Suite is a widely accepted and comprehensive statistical test set by the National Institute of Standards and Technology (NIST). The suite consists of 15 tests, each targeting a specific aspect of randomness. Key Randomness Analysis using the NIST test suite provides a detailed evaluation of the key's uniformity, independence, and unpredictability. The results of these tests offer valuable insights into the overall security of the cryptographic system, highlighting potential weaknesses or vulnerabilities due to insufficient randomness [53], [54], [55].

As the encryption/decryption process depends essentially on the XOR operation, the secret to ensuring the robustness of the proposed cryptographic algorithm is using random numbers generated by the Collatz Conjecture algorithm. Even if the input private key is tiny, 15-decimal digits, the confidentiality of the algorithm depends on the randomness degree of the generated random number sequences. These sequences were examined against several randomness tests to verify this assumption. Conventional run tests by Matlab were examined first. Besides this baseline test, 11 randomness tests proposed by NIST [53], and in [59] were examined too. To assert examination results, 10 sequences were examined by each test. Table 5 shows the results attained by each test. As indicated by the NIST testing suite [53], if eight out of the 10 input sequences pass the test, the random number generator succeeds the test. This characteristic provides extra novelty to the proposed methodology. The communicating parties can change the private key as they wish during their communication and still guarantee the robustness of the encrypted images.

TABLE 5. NIST test results for the generated key.

Test	Pass/Fail	Proportion	P-value
Approximate Entropy	Pass	(10/10)	0.405161
Block Frequency	Pass	(8/10)	0.101331
Cumulative Sums	Pass	(9/10)	0.323
Fast Fourier Transform	Pass	(10/10)	0.745603
Frequency	Pass	(9/10)	0.438578
Linear Complexity	Pass	(10/10)	0.947654
Overlapping Template	Pass	(10/10)	0.886589
Rank	Pass	(10/10)	0.039105
Runs	Pass	(9/10)	0.562559
Serial	Pass	(9/10)	0.778801
Non-Overlapping Template	Pass	(10/10)	0.999993

2) MATLAB RANDOMNESS TEST RESULTS

The Matlab Randomness Test Suite is a more specialized set of tests designed for the Matlab software environment. This suite offers several statistical tests and visualization tools that can be used to analyze key randomness. With its user-friendly interface and customizable options, the Matlab Randomness Test Suite allows for a more tailored and interactive analysis of key randomness, making it an invaluable tool for researchers and practitioners alike [56], [57], [58].

The Matlab randomness test was also performed on all the sequences. There are nine million possible numbers for a private key of size 15, in which 7 digits are used to generate random sequences after running through the Collatz Conjecture. All the 9 million are run through the Collatz Conjecture. Table 6 lists the average length of these random sequences, which is 635.68, and on average, 95% of these sequences pass the Matlab randomness test.

TABLE 6. Matlab randomness test results.

Sequence	Average length	Passed Matlab randomness test	Percentage
1st million	586.991192	962602	96%
2nd million	608.64514	956252	96%
3rd million	622.888048	951905	95%
4th million	633.490488	948730	95%
5th million	641.313192	945396	95%
6th million	647.912596	943539	94%
7th million	655.360012	941582	94%
8th million	659.52574	939173	94%
9th million	665.035248	938013	94%
Overall Average	**635.6846284**	**947465.78**	**95%**

B. KEY SENSITIVITY

To prove the sensitivity of the chaotic approach used in the proposed method, an experiment that studies the effect of changing a single digit in the private key on the quality of the

decrypted image is conducted. The results in Table 7 suggest that a slight change in a digit in the private key that is used for encryption and decryption will cause the recovered image to be drastically damaged, and this concurs with the fact that the proposed method is sensitive to initial condition and any slight change in the private key causes the original image to be lost and is considered a hacking attempt.

TABLE 7. Key sensitivity analysis.

Private key	MSE	PSNR	SSIM
'965412038794564' Original	0	Infinity	1
'965412038794563'	10170.0	18.5529	0.0002
'966412038794564'	10118.0	18.6050	0.0016
'963412038794564'	10156.0	18.5668	-0.0004

Figure 7 illustrates the significant impact of altering just one digit in the private key during decryption. Figure 7 consists of three images: (1) the original image, (2) the encrypted image, and (3) the wrongly decrypted image caused by the single-digit change in the private key (LSD changed from 4 to 3).



FIGURE 7. impact of single-digit change in Private Key on the decrypted image.

C. KEY SPACE

It can be observed from subsection IV-A that t has $10^7 - 10^6$ different values. p and q are allowed to be between 0 and 620 due to the average length X derived from Collatz Conjecture of t (or its derivatives $t + p, \ldots$). s is a two-digit number, but is used as an offset of p and q for indexing purposes.

Thus, the brute search space is:

$$(|t| \times |p|)^3 = (9E6 \times 620)^3 \approx 1.737E29$$
$$\text{which is slightly greater than } 2^{97}$$

. Thus, the search space for the brute-force attack is slightly below 2^{100}.

D. IMAGE QUALITY ANALYSIS

Mean Square Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Uniform Average Change Intensity (UACI), and Structural Similarity Index (SSIM) are commonly used metrics in image processing, including image encryption.

MSE measures the average squared difference between the original and encrypted images. A low MSE value indicates that the encrypted image is similar to the original image. However, MSE needs to provide more information on the perceptual quality of the image.

PSNR is a measure of the ratio between the maximum possible power of a signal and the power of corrupting noise that affects the fidelity of its representation. It measures the ratio of the maximum pixel value of the image to the root-mean-square error (RMSE) between the original and encrypted image. A higher PSNR value indicates that the encrypted image is more similar to the original image. In image encryption, these metrics can evaluate the encrypted image's quality and the encryption algorithm's effectiveness in preserving the image content. For example, a good image encryption algorithm should produce a low MSE value and a high PSNR value for the decrypted image. These metrics can also be used to compare different image encryption algorithms and to choose the most suitable one for a given application. They can be calculated using Equations (4), and (5) [28].

$$MSE = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}(x_{ij} - y_{ij})^2 \tag{4}$$

where m is the number of rows in the cover image, n is the number of columns in the cover image, x_{ij} is the pixel value from the original image, and y_{ij} is the pixel value from sent image.

$$PSNR = 10\log_{10}\frac{[MAX_I]^2}{MSE_t} \tag{5}$$

where MAX_I is the maximum signal value in our original "known to be good" image.

UACI is a crucial metric used in image encryption to evaluate the efficiency of encryption algorithms in terms of pixel intensity alteration. UACI is defined as shown in Equation (6), (where M and N are the dimension of the two images to be compared I and K) and the Number of Pixels Change Rate (NPCR), for which the mathematical formula is shown in Equation (7) [28]:

$$UACI = \frac{1}{M \times N}\left[\sum_{i,j}\left[\frac{|I(i,j) - K(i,j)|}{255}\right]\right] \times 100\% \tag{6}$$

$$NPCR = \frac{\sum_{i,j} D(i,j)}{M \times N} \times 100\% \tag{7}$$

where

$$D(i,j) = \begin{cases} 0 & \text{if } I(i,j) = K(i,j) \\ 1 & \text{otherwise} \end{cases}$$

In image encryption, UACI is a valuable tool for researchers and practitioners to assess the performance of encryption techniques and fine-tune their algorithms, ensuring optimal security and robustness against attacks [60]. By quantifying the average intensity change in pixel values between the original image and its encrypted counterpart, UACI provides insight into the uniformity of pixel distribution in the encrypted image. A higher UACI value indicates

that the encryption algorithm has effectively dispersed the pixel intensities, resulting in a more secure encrypted image.

SSIM is a widely used metric in image processing for assessing the quality of image encryption techniques by comparing the degree of structural similarity between an original image and its encrypted counterpart. Unlike conventional metrics like MSE or PSNR, which focus primarily on pixel-level differences, SSIM considers the human visual system's perception of image quality by considering luminance, contrast, and structural information. In image encryption, an effective algorithm should generate images visually dissimilar from the original, with no discernible patterns or correlations. SSIM is an essential tool for evaluating the performance of encryption algorithms by quantifying the degree of structural similarity between the original and encrypted images. A low SSIM value implies that the encrypted image has effectively concealed the original's structural information, making it difficult for potential attackers to recover the source image or gain insights from the encrypted data. Thus, SSIM plays a crucial role in developing and assessing image encryption techniques, guiding researchers and practitioners in optimizing their algorithms for enhanced security and data privacy. SSIM can be calculated according to Equation (8) [61].

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2\mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \tag{8}$$

where μ_x is the pixel sample mean of image x, μ_y is the pixel sample mean of image y, σ_x^2 is the variance of image x, σ_y^2 is the variance of image y, σ_{xy} is the covariance of image x and image y, $c_1 = (k_1L)^2$, $c_2 = (k_2L)^2$ are two variables used to stabilize division with a weak denominator, and L is the dynamic range of the pixel-value ($2^{\text{bits per pixel}} - 1$); here, by default $k_1 = 0.01$ and $k_2 = 0.03$.

Table 8 presents two essential quality parameters, MSE and PSNR, for evaluating the encryption method's performance. A higher MSE value indicates better encryption performance, implying a higher difference between the original and encrypted images. A lower PSNR value means the encrypted image has more noise or distortion than the original unencrypted image. A lower PSNR value indicates a more significant difference between the original and encrypted images, indicating a lower-quality image. Therefore, a lower PSNR value for encryption suggests that the encryption algorithm has introduced more noise or distortion into the image, making it more difficult to recover the original image.

Table 9 presents image quality results for the proposed encryption method, specifically focusing on SSIM, UACI, and NPCR metrics. Table 9 showcases the performance of the encryption method using these image quality metrics, offering insights into the method's effectiveness and ability to withstand potential attacks.

E. HISTOGRAM ANALYSIS

Histogram analysis is a technique used in image encryption to assess the encrypted image's statistical properties and compare them with the properties of the original image, such as pixel distribution and intensity levels. This analysis is essential for evaluating the robustness and security of image encryption algorithms, ensuring the encrypted images resist various attacks and exhibit uniform distribution and high entropy levels. Recent advancements in image encryption, such as the application of chaotic maps and deep learning techniques, have significantly improved the security and efficiency of encrypted images [62]. Furthermore, these novel approaches have successfully addressed the issues of pixel correlation and histogram analysis, making it increasingly difficult for potential attackers to exploit any statistical patterns in the encrypted images [63].

Figure 8 displays the histogram of the original image and its encrypted counterpart. In the encrypted image, the pixel distribution is more uniform, leading to a near-equal probability of occurrence for each intensity level.

F. CHI-SQUARE TEST

The Chi-Square Test is a statistical method used to determine the uniformity of data distribution and the independence of variables in image encryption systems. By analyzing the pixel

TABLE 8. MSE and PSNR results (between the source and encrypted images).

Kodak Image Index	MSE	PSNR
1	7686.6	21.3529
3	8532.4	20.309
4	8570.6	20.2643
6	9062.2	19.7066
7	7696.5	21.34
8	9391.9	19.3492
11	8658.8	20.1619
13	9154.2	19.6056
16	7967.1	20.9945
18	10702	18.0433
20	15253	14.5002
24	8890.5	19.8979

TABLE 9. Image quality measurements.

	Between input image and encrypted image			Between input image and recovered image
Kodak image index	NPCR	UACI	SSIM	SSIM
1	0.9961	0.2848	0.0004	1
3	0.9960	0.2977	0.0010	1
4	0.9960	0.2983	0.0015	1
6	0.9960	0.306	0.0003	1
7	0.9961	0.2849	0.0004	1
8	0.9961	0.3111	0.0004	1
11	0.9959	0.2997	0.0003	1
13	0.9962	0.3076	-0.0006	1
16	0.9961	0.2893	0.0004	1
18	0.9961	0.3306	0.0009	1
20	0.9961	0.4009	-0.0001	1
24	0.9962	0.3033	0.0001	1

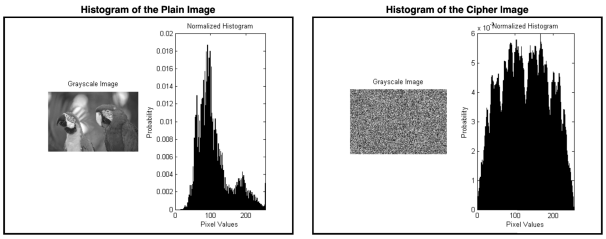


FIGURE 8. Histogram analysis.

value frequency distribution and comparing it to the expected uniform distribution, the Chi-Square Test can effectively measure the level of randomness and security in encrypted images [64]. It is beneficial in evaluating encryption schemes such as chaos-based and DNA-based encryption, where the unpredictability and diffusion of pixel values are vital for robust security. A lower Chi-Square value signifies higher randomness and improved encryption quality, indicating that the encryption technique effectively conceals the statistical properties of the original image [65].

In Table 10, the Chi-Square test results are presented as a quantitative measure of the effectiveness of an encryption method by comparing the distribution of pixel intensities between the original and encrypted images. Table 10 shows a significant difference between the distribution of the original and encrypted images, while the values of the encrypted images are smaller than the original images. This means a secure encryption method is used to encrypt the images.

TABLE 10. Chi-square test results.

Kodak Image Index	Plain Image	Cipher Image
1	517990	293.97
3	566930	283.57
4	502180	261.82
6	739390	261.34
7	515940	298.55
8	285280	224.19
11	766600	278.06
13	309170	282.12
16	434290	245.8867
18	393030	286.10
20	24420000	265.97
24	817770	219.99

G. INFORMATION ENTROPY ANALYSIS

Information entropy analysis is an essential aspect of evaluating the security and effectiveness of image encryption schemes. Entropy measures the randomness or uncertainty associated with a set of data. In the context of image encryption, it is used to assess the degree of randomness in the encrypted images. A high entropy value signifies that the encrypted image has a uniform distribution of pixel intensities, making it more challenging for an attacker to decipher or predict any patterns in the image. This, in turn, indicates that the encryption algorithm successfully conceals the original image's information and provides robust security against various attacks [66], [67], [68], [69], [70], [71].

Performing information entropy analysis on image encryption involves calculating the entropy of the original and encrypted images and comparing their values. An effective encryption scheme should yield an encrypted image with entropy close to the maximum possible value for 8-bit images. In practice, encryption algorithms that produce encrypted images with entropy values close to this maximum threshold are considered highly secure and resistant to attacks; the formula to calculate the information entropy is given in Equation (9) [66], [67], [68], [69], [70], [71].

$$H(I) = -\sum_{i=0}^{255} p(i) \lg p(i) \tag{9}$$

where $H(I)$ represents the information entropy of the image. $p(i)$ is the probability of occurrence of pixel intensity i in the image. The summation runs from 0 to 255, the possible pixel intensities for an 8-bit grayscale image.

Table 11 presents the entropy values for plain and encrypted images. The mean average for the plain and encrypted images is 7.1151 and 7.8879, respectively. As seen in Table 11, the encrypted image entropy values are notably more significant than those of the plain images, suggesting that the encryption procedure has added substantial randomness and enhanced the security of the resulting encrypted image.

TABLE 11. Information entropy results.

kodak Image Index	Plain Image Entropy	Cipher Image Entropy
1	7.1610	7.8931
3	7.0919	7.8623
4	7.1192	7.8049
6	7.4580	7.8903
7	7.0244	7.8861
8	7.6222	7.9166
11	6.8570	7.8639
13	7.4330	7.8784
16	7.1884	7.9149
18	6.9755	7.8998
20	6.3357	7.9377
24	7.1149	7.9065
Average	**7.1151**	**7.8879**

Table 12 shows a comparative analysis of information entropy among different algorithms, including the proposed and six reference algorithms. The proposed algorithm has an information entropy of 7.8879, comparable to other algorithms in the literature. The information entropy values for each algorithm are presented in the table.

H. CORRELATION ANALYSIS

The correlation Coefficient (CC) measures the linear correlation between two images. It measures how well the encrypted image's pixel values match the original image's. A high CC value indicates that the encrypted image is highly correlated

TABLE 12. Information entropy comparisons with other methods.

Method	Information Entropy
Xu et al. [66]	7.9893
Huang et al. [67]	7.9974
Lai et al. [68]	7.9976
Liu et al. [69]	7.9976
Zhou et al. [70]	7.9973
Qian et al. [71]	7.9026
Proposed Method	7.8879

with the original image; CC can be calculated as shown in Equation (10) [28].

$$CC(X,Y) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}} \quad (10)$$

where CC is the Correlation Coefficient, X_i is the value of the first message, $\bar{X}$ is the mean of X, Y_i is the value of the second message, and $\bar{Y}$ is the mean of Y, n is the total number of pixels in the image.

Table 13 presents the computed CC between the original and its encrypted counterpart utilizing the suggested method. The exceedingly low CC values between these images demonstrate a lack of any significant relationship between them.

TABLE 13. Obtained correlation coefficients.

Kodak Image Index	CC Red	CC Green	CC Blue
1	-0.0014	-0.0016	-0.0012
3	0.0005	-0.0019	0.0012
4	0.0002	-0.0008	-0.0011
6	-0.0004	-0.0003	-0.0006
7	-0.0003	-0.0004	0.0005
8	-0.0014	-0.0009	0.0001
11	0.0032	-0.0008	0.0017
13	-0.0003	-0.0017	-0.0015
16	-0.0022	-0.0028	-0.0028
18	0.0017	0.0004	0.0024
20	0.0005	0.0002	0.0004
24	-0.0039	-0.0033	-0.0032

Figure 9 shows a sample output where the calculated CC value between two image files expresses the dependency between their corresponding gray values.

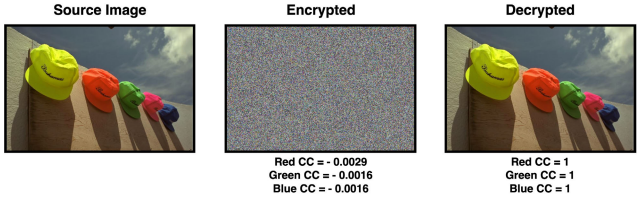


FIGURE 9. CC for encrypted and decrypted images.

Table 14 lists the correlation coefficients for horizontally, vertically, and diagonally adjacent pixels across various plain images and their ciphered counterparts. Correlation

TABLE 14. Image correlation coefficients.

kodak	Original Image			Cipher Image		
Image Index	Diagonal	Vertical	Horizontal	Diagonal	Vertical	Horizontal
1	0.7708	0.8354	0.8854	-0.0014	0.002	-0.0021
3	0.9642	0.9631	0.9807	-0.0009	0.003	0.0004
4	0.9481	0.9667	0.9633	0.0013	0.0007	0.0028
6	0.9232	0.9378	0.9754	0.0005	-0.0035	0.0003
7	0.9173	0.9416	0.9631	0.0013	0.0013	0.0022
8	0.8105	0.9198	0.8835	0.001	0.0015	-0.0032
11	0.893	0.9237	0.9431	-0.0017	0.0019	0.0001
13	0.809	0.8398	0.8961	-0.0019	0.0024	-0.0019
16	0.9376	0.9429	0.9835	-0.0007	0.0012	0.0011
18	0.8422	0.902	0.8875	-0.0018	0.0005	0.0007
20	0.9813	0.9871	0.9906	-0.0023	-0.0012	-0.0024
24	0.9039	0.9272	0.9378	0.0006	0.0001	0.0021

coefficients range between −1 and 1, with a coefficient of 1 signifying an ideal correlation between neighboring pixel pairs and a coefficient of 0, denoting no correlation. Increased correlation coefficients imply a strong correlation between adjacent pixel pairs, reducing the pixel distribution's randomness. The table illustrates that various correlation coefficients are high for original images and drastically low for ciphered images, increasing the difference and randomness between the two-pair images.

Figure 10 demonstrates the horizontal, vertical, and diagonal associations between adjacent pixels in the unencrypted and encrypted images using the outlined encryption method. Figure 10 also highlights the correlation coefficients for different directions before and after encryption. As evident in Figure 10, the correlation among adjacent pixels in the encrypted image is sparse and nearly linear in the input color image. This implies that the encrypted image exhibits a low correlation among neighboring pixels, making it more difficult for an attacker to decipher the original image. As a result, the proposed method offers high resistance against various attacks.

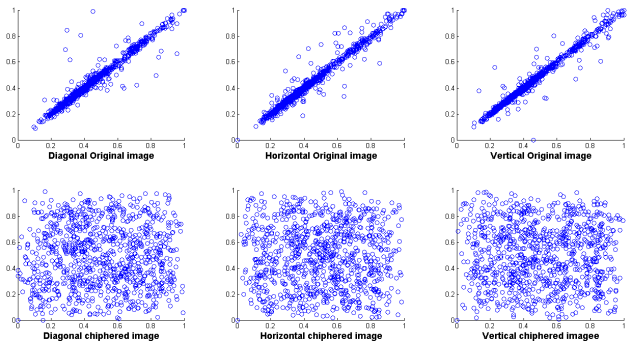


FIGURE 10. The correlation coefficient analysis.

I. SPEED ANALYSIS

To prove the proposed method's efficiency in terms of encryption time and throughput, the attained performance results are listed in Table 15. The average throughput is 0.4606 Mbps, equal to 471.6 Kbps, which is reflected in Table 16 compared with other state-of-the-art methods. At the same time, the average encryption time is 2.4444 seconds.

TABLE 15. Encryption time and throughput for the selected images.

kodak Image Index	Total Encryption Time (Seconds)	Throughput (Mbps)
1	2.4272	0.4635
3	2.5744	0.4392
4	2.4435	0.4605
6	2.4025	0.4683
7	2.4238	0.4642
8	2.4178	0.4653
11	2.4466	0.4598
13	2.4438	0.4604
16	2.4412	0.4609
18	2.4555	0.4585
20	2.4279	0.4634
24	2.4282	0.4633
Average	**2.4444**	**0.4606**

Table 16 presents information on the throughput enhancement achieved by applying the proposed method compared to other methods, including Non-Chaotic, Chaotic, and Hyper Chaotic approaches. Table 16 shows that the proposed method outperformed a previously proposed one in [28] by 1.56 times and some Non-Chaotic and Chaotic approaches. However, Hyper Chaotic approaches still generate a higher throughput than the proposed method.

TABLE 16. Throughput enhancement.

Method	Throughput (Kbps)	Abu-Faraj et al. [28]	Proposed Method
Non-Chaotic Approach [31]	170.3906	1.78	2.77
Chaotic Approach [31]	141.2305	2.14	**3.34**
Hyper Chaotic Approach [31]	636.3379	0.48	0.74
Yepioda et al. [36]	888.8867	0.34	0.53
Hua et al. [32]	638.4082	0.47	0.74
Asgari-chenaphlu [33]	911.0352	0.33	0.52
Zhang and Wang [34]	360.4102	0.84	1.31
Zhenjun and Sun [35]	384.9609	0.79	1.23
Abu-Faraj et al. [28]	302.7968	1.00	**1.56**
Proposed Method	471.6544	0.64	1.00

Table 16 shows the results of this comparison, indicating that the proposed method can achieve higher throughput than some of the Non-Chaotic and Chaotic approaches but is less effective than the Hyper Chaotic approaches. These results provide valuable insights into the effectiveness of different approaches for enhancing throughput and can help guide the selection of appropriate methods for specific applications.

Another experiment was performed to study the effect of the size of the 2D logistic map on the performance of the proposed method in terms of encryption time and throughput. Based on Table 17, we can deduce the following points:

- The size of the generated Chaotic logistic map used for encryption and decryption is variable, making the proposed method flexible and able to work on any image size without any problems.
- The number of rows and columns determines the size of the Chaotic logistic map, which is the key used for encryption as indicated in Equation (3).
- The map size can be either square or rectangular, i.e., the number of rows can be different from the number of columns.
- From the table, as the size of the key increases, the required time for encryption and the throughput are affected accordingly. However, the key size of 600×600 seems to be the most efficient regarding the required time for encryption and the maximum throughput. This is because most of the images in the Kodak dataset are of the size 512×768, making it faster when resizing the generated chaotic logistic key to match the image size to be encrypted. This also concurs with the previous finding of the Abu-Faraj et al. in [28].

TABLE 17. Encryption time and throughput for various private key sizes.

Row	Column	Total Encryption Time (Seconds)	Throughput (Mbps)
60	60	3.9681	0.2903
60	80	4.8485	0.2502
80	80	4.7514	0.2545
100	150	3.6444	0.3088
150	150	3.6082	0.3119
150	100	3.6082	0.3118
200	200	3.5252	0.3192
400	400	4.7405	0.2626
600	**600**	**3.0837**	**0.3719**
600	800	4.1103	0.2738
1000	1000	6.2758	0.1960
Average		**4.1967**	**0.2864**

J. SPECIAL PLAINTEXT IMAGE ATTACKS

A new experiment uses the proposed encryption scheme to encrypt plain black-and-white images. We select special images as input images, and the size of images for testing is $(512 \times 768 \times 3)$, which is similar to the image size in the used dataset. In Figure 11, Figure 12, and Table 18, the results show that the cipher image's information entropy increases significantly concerning the plaintext image in both cases, with a value as high as 7.94761, which is very close to the theoretical value.



FIGURE 11. All black image encryption using the proposed method.

K. NOISE ATTACK ROBUSTNESS ANALYSIS

Robustness to various noise types and data loss during transmission is a crucial evaluation process for image processing systems to determine their resilience against noise

TABLE 18. Special plaintext images results.

	Encrypted image						Decrypted image					
	MSE	PSNR	UACI	NPCR	SSIM	Information Entropy	MSE	PSNR	UACI	NPCR	SSIM	Information Entropy
All black image	2.2734e+04	10.5091	0.4987	0.9951	9.4544e-16	7.947610	0	NaN	0	0	1	-0.0000
All white image	2.2480e+04	10.6214	0.4948	0.9951	2.3269e-09	7.947610	0	Inf	0	0	1	-0.0000

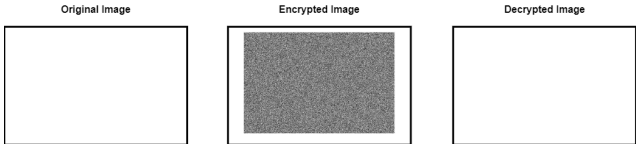


FIGURE 12. All white image encryption using the proposed method.

and distortion, such as salt-and-pepper noise, Gaussian noise, occlusion, and clipping.

1) SALT-AND-PEPPER NOISE

Salt-and-pepper noise, characterized by random white and black pixels in an image, can severely degrade visual information quality and hinder image cryptographic algorithms' performance. To assess the robustness of a cryptographic image system against this type of noise, researchers typically introduce varying levels of salt-and-pepper noise into the images and analyze the system's capability to mitigate or compensate for the noise-induced distortions. Various noise reduction techniques, such as median, mean, and adaptive filtering, can be employed and compared to evaluate their effectiveness in dealing with salt-and-pepper noise. Performance metrics like MSE and PSNR are often utilized to measure the system's robustness under different noise conditions quantitatively. By conducting a thorough robustness attack analysis, researchers and practitioners can identify potential weaknesses in cryptographic image systems and develop more reliable and noise-resistant algorithms, ensuring accurate and consistent performance in real-world applications [72].

Table 19 calculates and displays the PSNR and MSE values between the encrypted after adding noise and original images. When the noise level increases, the PSNR value decreases while the MSE value increases. A higher PSNR value indicates less loss of digital information, meaning that the original image can be more accurately reconstructed. On the other hand, a lower MSE value indicates less error or distortion between the original and the noisy image. Based on the results in Table 19, the proposed encryption system can effectively counter a salt-and-pepper noise attack even with significant noise intensity while maintaining the decrypted images' quality.

Figure 13 displays the results of adding Salt-and-Pepper noise with different intensities to an encrypted image, which enables the retrieval of the original image. The decrypted images have a strong correlation, and comparisons show highly favorable results, indicating a strong relationship

TABLE 19. Salt-and-pepper noise attack resistance.

Noise Level	PSNR	MSE
0.5	26.2576	4706.8
0.15	38.2762	1415.0
0.05	49.2768	471.0
0.005	72.5482	46.0
0.0005	95.9248	4.4

between the decrypted and original images. Therefore, the proposed method effectively mitigates the impact of noise on the image and can be considered an anti-noise algorithm.

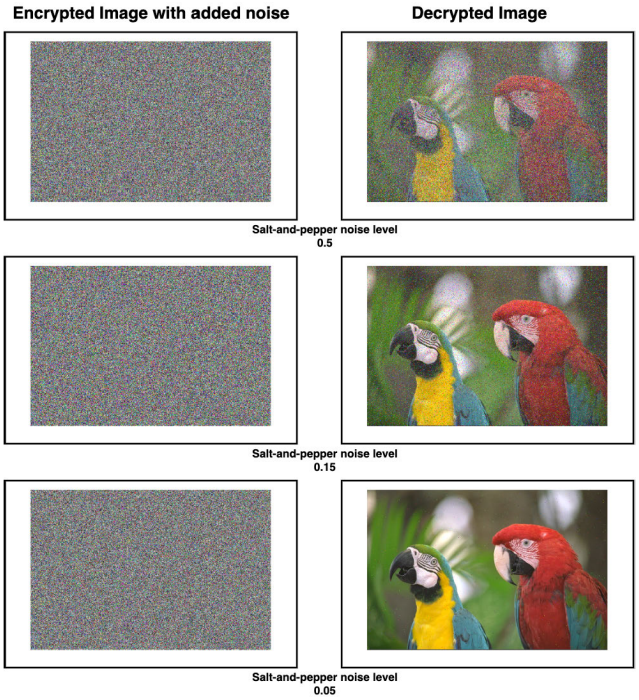


FIGURE 13. Robustness analysis of salt-and-pepper noise.

2) GAUSSIAN NOISE

Gaussian noise is a common type of random noise that can be added to images to simulate real-world conditions where images may be subjected to various disturbances. Table 20 shows Gaussian noise attack results for the proposed method. It evaluates the encryption method performance under varying levels of Gaussian noise. By comparing the MSE and PSNR values at different noise levels, one can

assess the robustness of the encryption method in preserving the quality of the encrypted images in the presence of noise.

TABLE 20. Gaussian noise attack resistance.

Noise Level	PSNR	MSE
0.5	19.8861	8901.0
0.15	27.6021	4114.7
0.05	32.3926	2548.5
0.005	33.4219	2299.2
0.0005	33.4406	2294.9

Figure 14 displays the results of introducing Gaussian noise of varying magnitudes to the encrypted image while still enabling the recovery of the original image. The decrypted images exhibit a high degree of similarity, and the comparison reveals highly favorable outcomes, indicating a robust relationship between the decrypted and original images. Consequently, it can be concluded that the suggested approach is Gaussian noise-resistant.

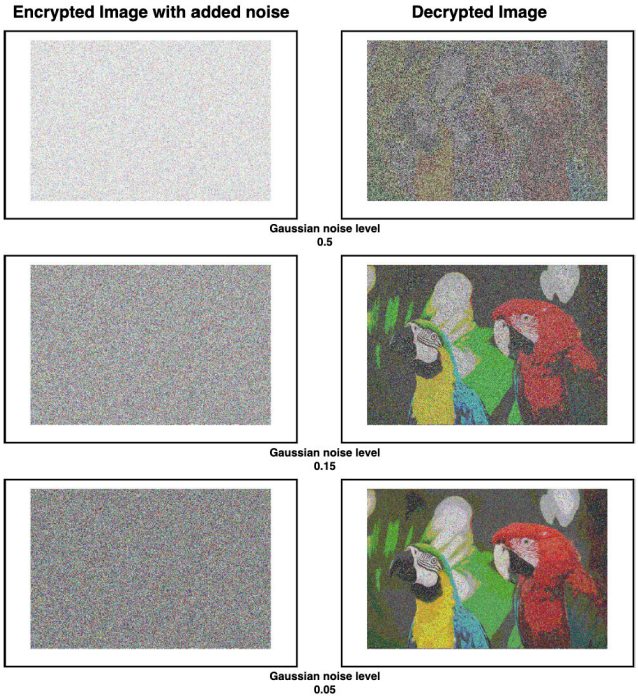


FIGURE 14. Robustness analysis of gaussian noise.

3) OCCLUSION ATTACKS

Occlusion attacks are a type of adversarial attack that can significantly impact the performance and security of image encryption systems. These attacks involve intentionally altering or corrupting specific regions within an image, often by overlaying patches, masking portions, or manipulating pixel values to conceal or distort the underlying information. Algorithms, in many cases, these attacks can lead to unauthorized access to sensitive information, misinterpretation of visual content, or even complete failure of the encryption system [73]. The primary goal of occlusion attacks is to exploit image encryption algorithms' vulnerabilities, compromising their confidentiality, integrity, and robustness.

Different levels of occlusion were examined, and the results are presented in Table 21, which illustrates the deterioration of various quality metrics (SSIM, MSE, and PSNR) as the percentage of occlusion increases.

TABLE 21. Occlusion attack analysis.

Occlusion %	SSIM-Decrypted	MSE-Decrypted	PSNR-Decrypted
0%	1	0.00	∞
5%	0.7052	547.30	47.78
10%	-0.5935	967.96	42.07
25%	0.3678	2377.60	33.09
50%	0.1837	4729.10	26.21
75%	0.0735	7062.70	22.20

Figure 15 and Figure 16 demonstrate that decrypted images obtained from encrypted images with 5%, 25%, and 50% occlusion are distorted but still recognizable to the human eye. However, decrypted images obtained from encrypted images with 75% occlusion are not recoverable. These results suggest that the proposed method can withstand occlusion attacks up to 50%.

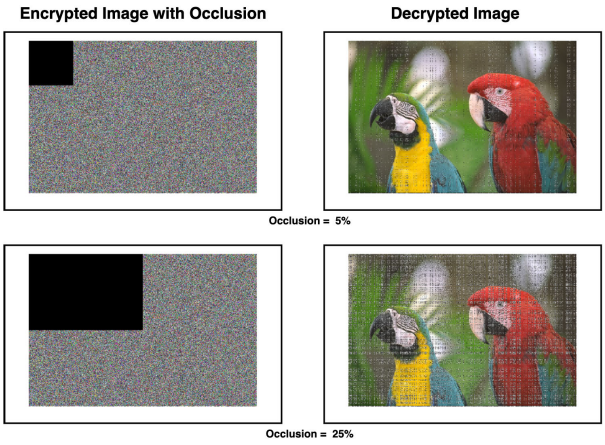


FIGURE 15. Low occlusion values.

4) CLIPPING ATTACKS

In a clipping attack, the attacker targets the image's dynamic range, altering the encrypted data's minimum and maximum pixel values. By doing so, the attacker can cause the image to lose contrast or change the overall appearance when decrypted. The goal of a clipping attack is to degrade the quality of the image, making it difficult for the recipient to understand or use the information in the image accurately [73].

From the results of various image quality metrics tested for the recovered image after the clipping attack in Table 22, it is clear that the proposed method is robust against clipping attack, especially for small changes in dynamic range values. Various metrics show degradation in their values when the

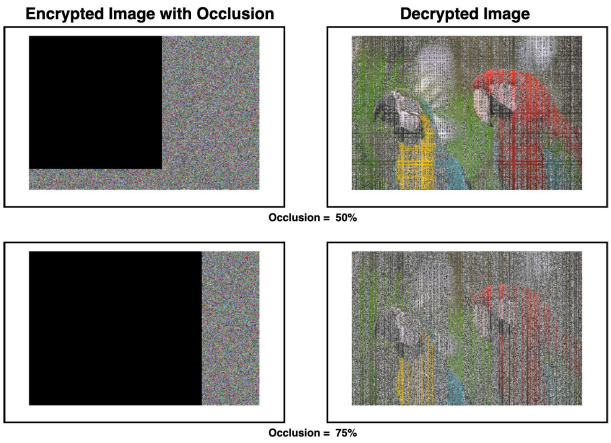


FIGURE 16. High occlusion values.

clipping range is increased. However, the human eye still understands the recovered image, as shown in Figure 17.

TABLE 22. Clipping attack analysis.

Clipping Range	MSE	PSNR	SSIM	NCPR	UACI
50-200	340.5246	52.5204	0.6534	0.4095	0.0398
25-225	42.9748	73.2191	0.8968	0.2142	0.0102
15-240	5.4282	93.9092	0.9812	0.1169	0.0026

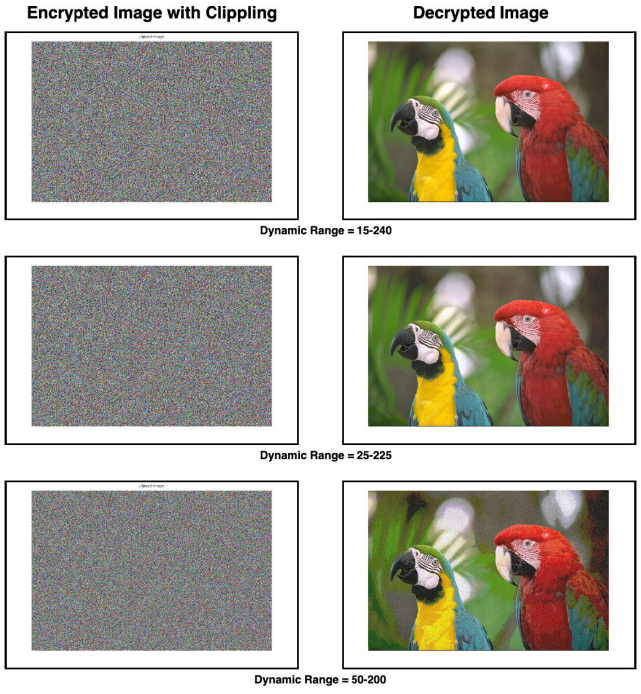


FIGURE 17. Different dynamic range values.

VI. CRYPTANALYSIS

The robustness of the proposed method against various attacks is discussed in this section. Observations about the performance of the proposed encryption scheme against common attacks are discussed in the following subsections:

A. CIPHERTEXT-ONLY ATTACKS

In ciphertext-only attacks, the attacker has access only to the ciphertext and no other information about the encryption algorithm or the encryption key. In the context of chaotic map encryption of images, the attacker only has access to the encrypted image and does not know the details of the chaotic map used for encryption or the specific encryption key.

Thus, in this case, since the research deals with the encryption of images, the attacker's goal is to gain knowledge about the original image or to find weaknesses in the encryption algorithm without having any information about the encryption process. The attacker's main approach in such attacks is to employ various cryptanalysis techniques to analyze the ciphertext and look for patterns or vulnerabilities to help break the encryption. Two of the ways these patterns or vulnerabilities may be analyzed would be to use:

- **Statistical Analysis:**
  The attacker may perform statistical analysis on the ciphertext to identify any statistical biases or regularities that could indicate weaknesses in the encryption scheme. They may look for pixel-value patterns, the correlation between neighboring pixels, or deviations from expected distributions. Correlation analysis that was conducted and discussed in Subsection V-H proved that there is no correlation between the original and its encrypted counterpart image, in Table 13 and Table 14, both tables list the small value of correlation in each color channel and in each direction of the encrypted image, shown in Figure 10, which means that there is no correlation between the encrypted image and the original image. Thus, it is difficult for the attacker to break such a system since no statistical information about the original images can be drawn from their encrypted versions.
- **Known-Plaintext Attacks:**
  Suppose the attacker obtains partial knowledge about the original image, such as a small portion of the plaintext and its corresponding ciphertext. In that case, they can attempt to exploit this information to deduce the properties of the encryption algorithm or the encryption key. Analyzing the known plaintext-ciphertext pairs may uncover patterns or relationships that help break the encryption. However, four rounds of random permutation are performed using random keys generated by Chaotic parameters; also, the spatial relationship between pixels has vanished, which is demonstrated by low correlation values in various directions of the encrypted images using the proposed scheme.

B. CHOSEN PLAINTEXT ATTACKS

In chosen plaintext attacks, the attacker has access to the cryptosystem's encryption machine, chooses special plaintext images, obtains their corresponding ciphertexts, and tries to find the encryption key. However, the original 15-digit

private key used for encryption is unrelated to the plaintext image, thus making it difficult to extract. Consequently, it is hard or almost impossible to extract any useful information about the encryption key. The proposed scheme is robust against this type of attack since it increases the entropy of the encrypted image compared to its plain counterpart, as was discussed in Subsection V-G. Moreover, the reduction of correlation coefficients of the plain image in the cipher image is evidence that the proposed scheme adds a substantial amount of randomness to the encrypted image, thus making it impossible to know.

C. CHOSEN CIPHERTEXT ATTACKS

The definition of a Chosen-ciphertext attack implies that the attacker can choose arbitrary ciphertext and have access to their corresponding plaintext decrypted from it since they have obtained temporary access to the decryption machine of the cryptosystem. According to [74], any authenticated encryption scheme is resistant to chosen-ciphertext attacks since it will not produce any message when decrypting.

D. DIFFERENTIAL ATTACKS

A differential attack is a cryptographic attack that breaks encryption schemes by analyzing the differences in input and output pairs. Image encryption refers to exploiting vulnerabilities in the encryption algorithm to reveal the underlying image or to find the encryption key. The primary goal is to recover the original image or gain unauthorized access to the encrypted data [36].

One main requirement of a cryptographic system to be robust against differential attack is that any small change in the plain image will cause the encrypted image to be greatly different from it. Two common metrics usually used to measure the difference between two images are UACI and NPCR, Equation (6), and (7), respectively. The measured values of these two metrics between original and encrypted images, as listed in Table 9, are very high and close to similar results reported in the literature. Thus, it indicates that the proposed method is robust against differential attacks. The Logistic Maps used to build the CLMK in this study are considered basic techniques in chaotic. Our goal next is to move to a more sophisticated and complex map of chaotic approaches other than CLM.

VII. CONCLUSION

This paper uses the Collatz Conjecture, a famous unsolved mathematical problem, to help generate 2D chaotic logistic maps' parameters for image encryption. The private key used in this paper is much shorter than the ones in the literature; only 15 decimal digits can easily fit in the Mantissa of a ''double'' data type. However, the novel way of parsing this short key with the help of the Collatz Conjecture and indexing technique generates random sequences that pass the NIST test of randomness, and it also generates a search space of 2^{97} for the brute-force attacks.

The 2D chaotic logistic maps generated using the proposed method are highly sensitive to initial conditions and private key contents, thereby preventing hacking attempts. An efficiency analysis was conducted to evaluate the image cryptography method's performance, showing that this method achieves a throughput enhancement of up to 3.34 more than other chaotic encryption methods in the literature. The robustness of the proposed method against the most common attacks was also proved. The proposed encryption scheme can be further extended by conducting more cryptanalysis to prove its robustness against well-known attacks, such as ciphertext-only attacks, known-plaintext attacks, and man-in-the-middle attacks. Furthermore, a closer inspection of the performance and throughput analysis of the proposed method needs to be conducted so that a major enhancement can be achieved compared to the state-of-the-art methods, especially since the proposed method deals with huge images that need more time to encrypt.

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